

OSSP PRACTICAL PROGRAMQ1: -

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that:

1. Accept a Linux command as input.
2. Create a child process using fork ().
3. Execute the command in the child process using an appropriate exec () system call.
4. Allow the parent process to wait for the child using wait.
5. Display the Process ID (PID) of both parent and child processes.

CODE: -

```
#include
<stdio.h>
#include
<stdlib.h>
#include
<unistd.h>
#include
<sys/types.h>
#include
<sys/wait.h> int
main() {

    char command[100];

    pid_t pid;

    printf("Enter a Linux
command: ");
    scanf("%99s", command);

    pid
    =
    fork
```

```

    ); if
    (pid
    < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }
    if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID: %d\n",
        getpid()); printf("Parent
        PID: %d\n", getppid());
        execlp(command, command,
        NULL);

        printf("Command execution f
        ailed!\n");

        exit(1);
    }
    else {
        printf("Parent Process\n");
        printf("Parent PID: %d\n",
        getpid()); printf("Child
        PID: %d\n", pid); wait(NULL);

        printf("Child process completed.\n");
    }
    return 0;
}

```

OUTPUT:-

```
[apple@apples-MacBook-Pro-2 OSSP % nano q1.c
```

```
[apple@apples-MacBook-Pro-2 OSSP % ./q1
```

```
Enter a Linux command: ls
```

```
Parent Process
```

```
Parent PID : 2917
```

```
Child PID : 2918
```

```
Child Process
```

```
Child PID : 2918
```

```
Parent PID: 2917
```

a.out	p3	p6	p9	q2.c
copy.txt	p3.c	p6.c	p9.c	sample.txt
p1	p4	p7	practical	skill
p1.c	p4.c	p7.c	q1	
p2	p5	p8	q1.c	
p2.c	p5.c	p8.c	q2	

```
Child process completed.
```